



Answers and Questions about Class Size: A Statewide Experiment

Author(s): Jeremy D. Finn and Charles M. Achilles

Source: *American Educational Research Journal*, Autumn, 1990, Vol. 27, No. 3 (Autumn, 1990), pp. 557-577

Published by: American Educational Research Association

Stable URL: <https://www.jstor.org/stable/1162936>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

American Educational Research Association is collaborating with JSTOR to digitize, preserve and extend access to *American Educational Research Journal*

Answers and Questions About Class Size: A Statewide Experiment

Jeremy D. Finn

State University of New York at Buffalo
and

Charles M. Achilles

University of North Carolina at Greensboro

A large-scale experiment is described in which kindergarten students and teachers were randomly assigned to small and large classes within each participating school. Students remained in these classes for 2 years. At the end of each grade they were measured in reading and mathematics by standardized and curriculum-based tests. The results are definitive: (a) a significant benefit accrues to students in reduced-size classes in both subject areas and (b) there is evidence that minority students in particular benefit from the smaller class environment, especially when curriculum-based tests are used as the learning criteria. A longitudinal analysis of a portion of the sample indicated that students in small classes outperform their peers in kindergarten classes of regular size and also gain more in reading outcomes during the second year. The question of why these effects are realized remains largely unanswered, but in light of these findings, is particularly important to pursue.

JEREMY D. FINN is a Professor of Education at the State University of New York at Buffalo, 408 Christopher Baldy Hall, Buffalo, NY 14260. He specializes in multivariate analysis and school and classroom processes.

CHARLES M. ACHILLES is a Professor of Education at the University of North Carolina at Greensboro, Greensboro, NC 27412-5001. He specializes in educational administration.

The question of class size continues to attract the attention of educational policymakers and researchers alike. To school personnel, small classes promise to facilitate increased student–teacher interaction, allow for thorough and continuous student evaluation, and provide greater flexibility in teaching strategies. Not only are these processes generally desirable, but they are exactly the processes that are likely to benefit students at risk for school failure (Slavin & Madden, 1987). Administratively, smaller classes reduce teachers’ responsibilities for paperwork and record keeping, allowing them to allocate more time to instructionally relevant activities. Under ideal circumstances, a small class may also minimize discipline problems because the teacher can more easily keep all students under a watchful eye. At the same time, the cost of smaller classes is high, requiring investment in both additional teacher salaries and additional classroom space. This investment is unlikely to be forthcoming unless the evidence that positive outcomes will be realized is substantial and consistent.

This is not the case. In 1978, Glass and Smith undertook an extensive meta-analysis of class-size research aimed at settling the issue unambiguously. The authors concluded without qualification “that reduced class-size can be expected to produce increased academic achievement” (p. iv). Instead of settling the issue, the report was met with criticism (some of it compelling) and extensive debate about the very existence, not to mention magnitude, of the “class-size effect.” Robinson and Wittebols (1986) also integrated a large number of empirical studies and arrived at a different set of conclusions:

Existing research findings do not support the contention that smaller classes will of themselves result in greater academic achievement of pupils. . . . There is research evidence that small classes are important to increased pupil achievement in reading and mathematics in the early primary grades. . . . There is evidence that pupils with lower academic ability tend to benefit more from smaller classes than do pupils with average ability. (pp. 18–19)

Slavin (1989) reviewed eight well-designed studies, concluding that substantial reductions in class size do indeed have modest benefits among young children. However, the author noted that long-term studies indicate that the effects are not cumulative across grades and may even disappear in later years. The controversy over the existence of an effect continues, as well as its magnitude and whether it is more likely to accrue to student groups identified by age, gender, race or ethnic origin, or ability.

Only randomized experiments can provide a definitive answer to these questions. Correlational studies, and studies that match children after they have been assigned to small or large classes, are confounded, at the very least, by the effects of differential selectivity of students or teachers, or both. The entire Glass and Smith meta-analysis included only five randomized

experiments of nontrivial duration in which large and small class sizes were within the range found in most elementary and secondary schools.¹ The ten effect sizes for small class–large class comparisons that are derived from these studies range from $-.29$ to $+.17$. Robinson and Wittebols (1986) make no distinction among studies by methodology. Slavin (1989) included only one randomized experiment in his review, a large-scale study in grades 4 and 5 in Toronto, Ontario (Shapson, Wright, Eason, & Fitzgerald, 1980). The overall effect size for the study was $+.17$, but the small class–large class differences were negligible for reading ($-.05$ to $+.01$) and large for mathematics ($+.22$ to $+.51$). The smallest class size, 16, showed the most distinct advantage over classes of 23, 30, or 37 students.

The present study builds on the findings of this earlier research. It begins in the earliest years of school and continues for several years; it includes children identified by race and socioeconomic levels; and it is a controlled, randomized experiment. The reduction in class size is substantial, from an average of about 22 students present and providing data in the regular classes, to about 15 in small classes, a decrease of 32%.

However, an important qualification must be recognized. In this study, as in the others, a class with 25 students registered does not mean that 25 students actually attend class, and certainly does not imply that 25 students are present on any given day. For a plethora of reasons, the actual number may be somewhat smaller, much smaller or, on occasion, larger than 25. The problem of definition is complicated further when class size is used as an independent variable in an empirical study. The number of students actually *involved* in instructional activities may be more important to many outcomes than the number present, and class enrollment is only an approximation of this difficult-to-measure construct. In spite of these considerations, most studies do not indicate how enrollment figures were obtained, or their stability over time. Caution must be exercised in interpreting the effects of different degrees of class-size reduction. Attempts to study fine graduations in enrollment figures in particular—for example, by calculating ordinary correlations of class size with other measures—should be viewed with skepticism.

Procedures

All school systems in the State of Tennessee were invited to participate, with the state funding the cost of extra teachers and aides that would be needed.² About one third of the districts, representing 180 schools, expressed an interest in participating. After negotiation, the final sample consisted of 76 elementary schools that were large enough to have at least three kindergarten classes. Comparison of the participating school districts in 1986 showed that they were larger than the state average, had slightly higher per-pupil expenditures and teacher salaries, but were very similar in teacher–pupil ratios and percentages of teachers with higher degrees. Proj-

ect schools scored slightly below the state average in second-grade reading and mathematics standardized test scores. Schools were classified as inner city, urban, suburban, or rural.

Within each school, children entering kindergarten were assigned at random, by the project staff, to one of three class types: small, with an enrollment range of 13–17 pupils; regular, with an enrollment of 22–25 pupils; or regular with aide, with 22–25 pupils but with a teacher aide formally assigned to work with the class. Teachers were assigned at random to classes as a separate step. Except for this intervention, the school was to continue to operate as usual within district policies; that is, the study did not dictate any operational changes. No special training was provided for the teachers to help them take better advantage of the experimental settings.³ The within-school design is particularly effective in controlling for contaminating sources of variability attributable to differences among school settings; these may include characteristics of the student population (e.g., socioeconomic status, community resources), of the school (e.g., availability of instructional materials, school administration), or of the district (e.g., policies, per-pupil expenditures). Larger schools allowed for more than one class of each type. The resulting sample consisted of 328 kindergarten classes and about 6,500 pupils.

Every class was to remain the same type—small, regular, or regular with aide—for 4 years beginning in the fall of 1985. The project staff randomly assigned a new teacher to the student group in each subsequent grade. The results reported here are for the end of Grade 1, the second year of the study.

Several operational complexities affected the composition of the sample between kindergarten and first grade. For one, kindergarten is not mandatory in the State of Tennessee. As a result, some youngsters did not attend kindergarten at all, and others were enrolled in nonproject kindergartens.⁴ Participants entering the study in first grade were also assigned to classes at random; the process was monitored carefully by the project staff. The resulting sample from first grade, even with the loss of students who moved away from project schools, was larger than in kindergarten. The total number of classes was 347, with about 7,100 students; 34% had not participated in the project the previous year. After the data were screened for coding errors, 6,570 students in 331 classes remained for analysis (122 small, 111 regular, and 98 regular with aide) with the number of students varying slightly from one measure to another.

Two, to alleviate some parental concerns, about half of the regular-class students in kindergarten were randomly reassigned to teacher-aide classes in first grade, and half of the teacher-aide pupils were reassigned to regular classes. Youngsters in small classes were not reassigned. Thus, the comparison of teacher-aide classes with regular classes is confounded

Table 1
Description of First Grade Cross-Sectional Sample

Quantity	Location			
	Inner-city	Urban	Suburban	Rural
Schools	15	8	15	38
Classes				
All white students	0	18	28	119
All minority students	65	0	13	0
Mixed	5	23	21	39
Total	70	41	62	158
Students	1495	804	1214	3059
Percentage receiving free lunch	91.2	46.6	33.2	42.8
Percentage of minorities	96.2	21.1	39.4	7.2

Note. Numbers of students vary slightly from measure to measure.

by the mixed composition of those groups in Grade 1. The comparison of classes by enrollment remains unconfounded.⁵

Information on race and a rough index of socioeconomic status (whether or not the individual participated in the subsidized lunch program) were obtained for each youngster.⁶ These data are summarized in Table 1. At the extreme, 65 of 70 inner-city classes have only minority students, whereas there are no all-minority classes in the urban or rural samples, and minorities are a small proportion in the suburbs. There is a strong association between minority status and participation in the free lunch program. About 70% of the student sample are either minorities receiving free lunches, or whites not receiving free lunches; Yule's *Q* association measure in .78. An attendance index, the proportion of school days that the individual was present in class, was also calculated for each student. The mean proportion, .95, did not differ systematically by class type, school location, or race.

Outcome Measures

The Stanford Achievement Tests (SATs) in reading and mathematics were administered in the spring of each school year. In addition, beginning in first grade, a set of curriculum-referenced tests developed by the State of Tennessee, the Basic Skills First (BSF) tests, was administered to each student (Tennessee Department of Education, 1987). These were constructed from well-specified lists of objectives in reading and mathematics at each grade level, and could be scored either as the total number of items answered correctly, or as pass-fail. A student passes if he or she masters 80% of the objectives covered by the test items. The pass-fail system was used in the analysis reported here.

The Self-Concept and Motivation Inventory (SCAMIN) was given to the classes at the end of each school year (Milchus, Farrah, & Reitz, 1968). This group-administered instrument yields a measure of academic self-concept and academic motivation for each student. Its questions and answers, keyed to pictures of common objects and faces that appear happy or sad, is appropriate for children in the primary grades. Its reliability for early grades is only moderate, with alpha coefficients between .56 and .69 for subscales (Davis & Johnston, 1987; Davis, Sellers, and Johnston, 1988). Nevertheless, it provides convenient and useful measures for group comparisons such as required by this project.

Analyses

Means on each of the outcome measures were calculated for each class, and then separately for white and minority students in each classroom.⁷ Two analyses were performed, a cross-sectional analysis of the entire first-grade sample, and a longitudinal analysis of the subset of pupils who were in the study for both kindergarten and first grade. Both analyses employed multivariate analysis of variance using the MULTIVARIANCE program (Finn & Bock, 1985).

The outcome measures for the cross-sectional analysis were considered in three sets: reading, comprising the SAT word study skills and reading subscales and the BSF reading test; mathematics, comprising the SAT total mathematics scale and the BSF mathematics test; and motivation/self concept, comprising the two SCAMIN subscales. The reading subscale of the SAT is a composite of both word reading and reading comprehension measures (The Psychological Corporation, 1985). Each BSF measure was scored as the percentage of students in the class who passed. The percentage was transformed into a log-odds index, $\ln(\%/100 - \%)$ for each class, to improve normality. Tests of significance for the class-type effect were accompanied by two contrasts, a comparison of small classes with the average of the other two class types, and a comparison of teacher-aide classes with regular classes.

The statistical design treated schools as a random dimension, nested within the four locations (inner city, urban, suburban, and rural). These were crossed with class type (small, regular, and aide) and race (white or minority). The mean scores for whites and minorities in each class were used as the basic units of analysis. The design has unequal *N*s and many empty cells, and required multiple error terms to test all of the fixed effects.

Of the three instruments used in the study, only the SATs permitted a comparison of scale scores across grade levels. The Stanford Early School Achievement Test 2 form was administered in kindergarten and Primary 1 in first grade. The longitudinal analysis was performed on scores from pupils who participated in the same experimental condition in the same

school location for 2 consecutive years—about 35% of the Grade-1 sample. Means were obtained on all three SAT scales—word study skills, reading, and total mathematics—for white and minority pupils in each class for both years. These became the data for a repeated-measures analysis of variance with grade as the within factor. The sample design was the same as the cross-sectional analysis previously described. The addition of the grade dimension permitted an examination of growth from kindergarten to first grade, and the interaction of growth with location, race, and class type. Multivariate tests of significance were obtained for the two reading scales jointly, and univariate tests were obtained for mathematics.

Cross-sectional Results

The analysis of variance results are summarized in Table 2 for the two sets of cognitive measures. This table, compiled from many computer runs, presents the multivariate test of each main effect and interaction, and separate univariate tests for each measure. Mean scores on the cognitive measures for the three class types, separated by location and race, are given in Tables 3 and 4, respectively.

Table 2
Analysis of Variance for Cognitive Outcomes

Effect ^a	Reading				Mathematics		
	Multivariate ^b	Word study skills	SAT reading	BSF reading	Multivariate ^b	SAT math	BSF math
Location	$p < .01$	$p < .01$	$p < .06$	—	$p < .05$	—	—
Race	$p < .001$	$p < .001$	$p < .001$	$p < .001$	$p < .001$	$p < .001$	$p < .001$
Type	$p < .001$	$p < .001$	$p < .001$	$p < .001$	$p < .001$	$p < .001$	$p < .05$
Location $\times$ Race	$p < .05$	—	—	$p < .05$	—	—	—
Location $\times$ Type	—	—	—	—	—	—	—
Race $\times$ Type	$p < .05$	—	$p < .05$	$p < .01$	—	—	—
Location $\times$ Race $\times$ Type	—	—	—	$p < .05$	—	—	$p < .01$

Note. Only statistically significant results presented; dash indicates statistically insignificant results.

^aThe nonorthogonal design necessitated tests of significance in several orders (Finn & Bock, 1985). The results presented here were obtained as follows: each main effect was tested eliminating both other main effects; Location $\times$ Race tested eliminating main effects and Location $\times$ Type; Location $\times$ Type tested eliminating main effects and Location $\times$ Race; Race $\times$ Type tested eliminating main effects and other two-way interactions; Location $\times$ Race $\times$ Type tested eliminating all else. ^bObtained from F approximation from Wilks' likelihood ratio.

Table 3
Mean Scores on Cognitive Outcomes by Location and Class Type

Group	Measure				
	Word study skills	SAT reading	BSF reading ^a	SAT math	BSF math ^a
Inner-City	497.9	486.7	51.7	510.7	71.1
Small	506.0	493.9	61.2	517.4	77.0
Regular	491.9	482.4	46.3	505.9	66.8
Aide	495.2	483.2	46.6	508.5	69.1
Urban	534.5	518.8	65.4	530.8	81.9
Small	544.9	531.1	74.9	540.5	84.7
Regular	527.0	508.3	61.0	526.5	82.9
Aide	532.4	518.0	60.6	525.6	77.6
Suburban	528.0	513.1	62.6	528.7	83.4
Small	540.9	523.9	68.9	540.8	89.0
Regular	518.7	507.4	58.6	521.1	80.4
Aide	523.5	506.5	59.8	522.8	79.9
Rural	537.0	521.6	63.3	536.7	83.4
Small	541.8	528.9	68.3	541.6	85.1
Regular	531.9	512.9	57.0	531.8	80.1
Aide	537.0	522.5	64.2	536.2	84.7
Standard deviation ^b	51.55	55.48	48.56	41.60	38.91

^aBSF results tabled are average percentage passing. Statistical analyses were performed using the log-odds transformation. Standard deviations obtained by scoring each pupil 0 (fail) or 1 (pass). ^bStandard deviations for all pupils in regular classes only.

The effects of location, race, and their interaction help to describe the context in which the experiment was conducted. Minorities score significantly below whites on all of the cognitive measures. There are significant differences among school locations, but the distinctions are not exhibited consistently across reading and mathematics subscales. However, when location differences are tested without eliminating the race main effect,⁸ both multivariate tests are significant at $p < .001$, and all of the univariate F ratios are significant with the exception of the BSF reading measure. That is, there are substantial differences among locations in average cognitive performance, but they are attributable in part to the racial compositions of the communities.

The interaction of location and race reflects very distinct minority-white differences in rates of passing the BSF reading test. The rural difference, on the average, is 20 percentage points—67.3% of whites pass compared with 47.1% of minorities. The suburban and urban differences are

Table 4
Mean Scores on Cognitive Outcomes by Race and Class Type

Group	Measure				
	Word study skills	SAT reading	BSF reading ^a	SAT math	BSF math ^a
White	540.2	524.8	66.5	540.1	86.2
Small	545.2	530.3	69.5	545.8	88.1
Regular	534.4	518.1	62.3	535.4	85.1
Aide	540.2	525.3	67.1	538.2	85.0
Minority	509.0	495.7	53.7	513.4	72.9
Small	518.6	507.1	65.4	521.3	77.8
Regular	503.5	489.0	48.0	509.2	69.4
Aide	505.6	491.7	48.3	510.3	72.2
Standard deviations ^b					
White	50.42	56.04	47.26	40.37	34.34
Minority	44.58	47.71	49.90	37.97	45.16

^aBSF results tabled are average percentage passing. Statistical analyses were performed using the log-odds transformation. Standard deviations obtained by scoring each pupil 0 (fail) or 1 (pass). ^bStandard deviations for all pupils in regular classes only.

3.6 and 8.3%, respectively. Likewise, although not statistically significant, the minority–white difference is the largest in rural schools on three of the four other cognitive measures. The pattern is reversed in inner-city schools, in which the average pass rate on the BSF reading test for whites is 46.9%, and 52.1% for minorities. Although the sample contains very few inner-city classes with white students, their performance is substantially below that of whites in other locations on every measure.

Do small classes provide an advantage to primary-grade children? The question is answered most directly by the class-type main effect. Comparing all three class types, there is a statistically significant effect for both multivariate sets and separately for each reading and mathematics measure. Further, Hotelling's T^2 for the comparison of small classes with the others is significant for both the reading and mathematics variable sets at $p < .001$; neither contrast of teacher-aide classes with regular classes is significant. The nonsignificant interaction with location confirms that the differences among class types occur to about the same extent regardless of the school setting. On the average, small classes positively affect student performance in first grade.

The mean scores for the three class types are given in Table 5 and are compared by effect sizes. The contrasts, small–others and aide–regular, are expressed in two different metrics. The first is the standard deviation of

Table 5
Means and Effect Sizes on Cognitive Outcomes
for Three Class Types

Class size	Measure				
	Word study skills	SAT reading	BSF reading ^a	SAT math	BSF math ^a
Means					
Small	535.5	521.8	68.0	536.9	84.4
Regular	520.6	505.1	55.9	523.7	78.1
Aide	527.2	512.7	60.0	527.7	80.2
Effect Sizes ^b					
Small-Others					
Student <i>SD</i> ^c	.22	.23	.21 (10.1%)	.27	.13 (5.2%)
Class <i>SD</i> ^d	.62	.64	.51	.66	.32
Aide-Regular					
Student <i>SD</i> ^c	.13	.14	.08 (4.1%)	.10	.05 (2.1%)
Class <i>SD</i> ^d	.35	.37	.26	.23	.15

^aBSF results tabled are average percent passing. Statistical analyses were performed using the log-odds transformation. Standard deviations obtained by scoring each pupil 0 (fail) or 1 (pass). ^bThese are differences in standard deviations. ^cStandard deviations for all pupils in regular classes only. ^dStandard deviations across all regular class means. BSF class-level effect sizes based on log-odds transformation.

the particular subscale across all youngsters in the study who attended regular classes; the second is the standard deviation computed among the means of all regular classes. Although the former are more traditional effect-size measures, decreasing class enrollment is a classroom-level intervention. It is important to ask, What is the impact on the class' average performance, relative to the usual distribution of classroom outcomes? In general, it can be observed that an increment of, say, *x* units is larger on the distribution of means than on the distribution of individual scores. This is correct. It reflects the fact that the effort required to raise a class average by that amount is generally much greater than that needed to augment one student's performance. For all effect sizes, standard deviations were obtained from regular classes only, as these were not impacted by the experimental interventions.

The benefit of the reduced class size is seen clearly for every measure. The effect sizes are about one fourth of a standard deviation among students, and range from about one third to two thirds of a standard deviation among class means. The difference in classroom pass rates on the BSF curriculum-based tests is 5.2% in mathematics and over 10% in reading. In fact, the univariate contrast values for comparing small classes with the others is significant at *p* < .001 for four of the five measures, and *p* < .01 for the

BSF mathematics scale. If the SAT means are referred to the publisher's table of norms, the small-class advantage is at least 1½ months for reading and 2½ months for mathematics on the grade equivalent scale.

The effects of augmenting a regular class with a full-time teacher aide are not as large as the small-class advantage, and are not statistically significant in the multivariate model. However, the average performance in teacher-aide classes is higher than regular classes on each measure. Tests of the comparison of aide classes with regular classes on each measure separately are significant for the three reading measures at $p < .05$, if one-tail tests are made. Although these tests are not statistically independent, they indicate that using teacher aides in the primary years may be helpful in augmenting reading performance. Further data are needed before this speculation can be confirmed. The experiment will continue through third grade with no further reassignments. By that time, data will be available on youngsters who have been in teacher-aide classes for 3 years, providing clearer answers to this aspect of the investigation.

Differential Effects on Whites and Minorities

The significant interactions of type with race indicate that, for some outcomes, reduced class size does not have the same impact on white and minority students. This can be seen clearly by examining the small-class advantage—the difference between the average performance of small classes and the average of all other classes (large and regular)—for white and minority students separately (see Table 6). For all five measures, the advantage of being in a small class is greater for minority students than for whites. For example, the small-class advantage for white students is 8.6 points (.15 σ) on the SAT reading scale. In contrast, minorities in small classes outperformed their peers by an average of 16.7 points (.35 σ), more than twice the effect size for whites.

Table 6
Small-Class Advantage by Race^a

Measure	White		Minority	
	Mean difference	Effect size ^b	Mean difference	Effect size ^b
Word study skills	7.9	.16	14.1	.32
SAT reading	8.6	.15	16.7	.35
BSF reading ^c	4.8%	.10	17.3%	.35
SAT math	9.0	.22	11.6	.31
BSF math ^c	3.1%	.09	7.0%	.16

^aEach value is Small – (Regular + Aide)/2; means from Table 4. ^bMean differences divided by the standard deviation for all white students or all minority students in regular classes. ^cBSF results tabled are average percentage passing. Statistical analyses were performed using the log-odds transformation.

The Race $\times$ Class size interaction effect on the curriculum-based BSF reading test is so powerful that the difference in pass rates between whites and minorities is reduced from 14.3% in regular classes to 4.1% in small classes (see Table 4). Minority students in small classes pass the BSF reading test at a rate generally comparable to their nonminority peers. The same pattern, although not statistically significant, is seen for word study skills and for both mathematics scales. The effect on mathematics may have been more dramatic if the teachers had devoted more instructional time to this subject. Each teacher completed a 1-day log indicating the number of 15-minute time intervals spent on reading and mathematics instruction. On the average, teachers reported that 9.2 time intervals (2.3 hours) were spent on reading, and 3.5 (.9 hours) on mathematics. Although these data may not be highly reliable, differences in time spent on the two school subjects are substantial and consistent across locations and class types.

Both scales of the BSF test give indications of a three-way Location $\times$ Race $\times$ Type interaction. To investigate, separate Race $\times$ Type analyses were conducted within each of the four locations, with these two measures as criterion variables. The interaction of race with class type was not statistically significant in urban or rural schools, although the small-class advantage in reading was greater for minorities than for whites in both locations. The interaction was statistically significant ($p < .01$) for BSF reading in both the inner-city and suburban schools.

There are very few classes with white students in the inner-city sample, so that a comparison between races may not be generalizable. However, for minorities the small-class pass rate is 64%, compared with 45.4% for the average of regular and teacher-aide classes. In the suburban schools, fully 74.7% of minorities in small classes passed the reading test, compared with 53.1% in other class types. The small-class advantage is small or nonexistent for suburban white students, whose average pass rate is 65.5% in small classes and 63.4% in the others.

The Race $\times$ Type interaction was also significant in the suburban sample for BSF mathematics at $p < .05$, and approached significance in the inner city ($p < .08$). The same pattern occurs once again. In both locations, the difference in pass rates between small classes and the other class types is more substantial for minorities than for whites. In fact, in the suburbs, the average percentage of minority students in small classes who pass the BSF mathematics test is 90.3. This is well above the pass rate for suburban minority students in other class types (70.7 %), and slightly above the average percentage of whites passing in *any* type of class in any location.

Motivation and Self-Concept

An analysis parallel to that performed for the cognitive measures was also applied to the motivation and self-concept scales. Only the location and race main effects reached statistical significance at $p < .05$, applying the

Table 7
**Mean Scores on Noncognitive Outcomes by
 Location, Race, and Class Type**

Group	Measure	
	Motivation	Self-concept
Location		
Inner-City	49.7	48.4
Urban	50.5	48.6
Suburban	49.7	47.9
Rural	50.2	47.7
White	50.2	47.7
Small	50.1	48.0
Regular	50.2	47.4
Aide	50.2	47.7
Minority	49.9	48.5
Small	49.9	48.6
Regular	49.9	48.5
Aide	49.9	48.3
Standard deviation	3.85	5.37

Note. Standard deviation is computed from all students in regular classes.

multivariate test criterion. Univariate results reveal that differences among locations are significant for motivation but not self-concept, with urban and rural schools having slightly higher average levels than inner-city or suburban schools (see Table 7). The difference between white and minority students is statistically significant for self-concept alone, at $p < .01$; minority students have higher average levels of self-concept as revealed on the SCAMIN instrument. Although similar race differences have been found in previous research (Wylie, 1979), they have not been documented among children as young as these.

Longitudinal Results

An analysis of 2-year longitudinal data was conducted for that subset of youngsters who participated in the same classroom arrangement in the same geographic region for both kindergarten and first grade, and who had complete data on the SAT achievement tests for both years.⁹ Again, classroom means were computed for white and minority students separately, and these were used as the base data for analysis of variance. As with the cross-sectional analysis, if a mean was based on two observations or fewer, it was excluded from the analysis. Due to the number of students who did not attend kindergarten, those who were exchanged between regular and teacher-aide classes, and these additional data requirements, the resulting longitudinal sample consisted of 2291 youngsters—about 35% of the cross-sectional sample.

Table 8
Description of Longitudinal Sample

Quantity	Location			
	Inner-city	Urban	Suburban	Rural
Schools	15	8	15	36
Classes	60	38	51	152
Students	448	257	336	1250
Percentage receiving free lunch	92.8	37.0	28.1	36.5
Percentage of minorities	91.1	8.3	36.0	3.0

The composition of the longitudinal sample is described in Table 8. Altogether, 301 classes provided data (113 small, 99 regular, and 78 teacher-aide) compared with 331 classes in the cross-sectional analysis. The longitudinal sample comprises a smaller percentage of minority students and, with the exception of the inner-city schools, a smaller percentage of pupils in the subsidized lunch program. This may reflect a greater tendency for poor, minority children to begin school in first grade, and thus not appear in the 2-year data set.

The longitudinal sample is superior to the total group in terms of standardized achievement scores. Comparison of the first-grade means (Table 9) with those of Table 5 shows that the longitudinal sample outperforms the larger group on all three SAT scales in all class types. It is likely that youngsters who attend kindergarten come from homes in which early schooling is given a high priority. In addition, they enjoy the advantage of an extra year's school work. However, the rank order of means among small, regular, and teacher-aide classes is the same in both samples on every measure.

The repeated-measures analysis of variance yields results for location differences, race differences, and differences among class types that are consistent with the cross-sectional findings (see Table 10). Differences among the four geographic locations and between minorities and whites are statistically significant for the pair of reading outcomes and for all three SAT scales individually. Differences among classes types are also statistically significant: Small classes have a clear advantage over the other class types on all scales in both grades. Neither the Location $\times$ Race nor the Location $\times$ Type interaction is significant, indicating that the race and class size effects are found in all four geographic areas.¹⁰

There was significant overall growth from kindergarten to the end of first grade on all three SAT scales. The mean differences between grades are large (see Table 9) and are also distinguished by location, race, and class type. That is, all three interactions are significant for the reading scales and Grade $\times$ Location interaction is significant for mathematics. Differential

Table 9
Mean Scores and 1-year Growth for Longitudinal Sample

Measure	Class type	All Students			White			Minority		
		K	1	Growth	K	1	Growth	K	1	Growth
Word study skills	Small	451.5	542.4	90.9	454.9	551.4	96.5	443.9	522.6	78.7
	Regular	444.1	525.0	80.9	447.6	538.3	90.7	436.8	497.3	60.4
	Aide	442.8	533.5	90.7	445.0	546.0	101.0	436.4	496.6	60.2
Reading	Small	442.4	529.5	87.1	446.1	536.6	90.5	434.3	514.2	79.8
	Regular	435.4	509.4	74.0	438.0	520.9	82.9	430.0	485.2	55.2
	Aide	435.1	520.3	85.2	436.9	530.9	94.0	429.8	489.0	59.2
Mathematics	Small	497.4	540.7	43.4	502.8	548.2	45.3	485.2	524.2	39.0
	Regular	486.8	528.1	41.3	489.4	538.5	49.1	481.5	507.0	25.5
	Aide	488.6	534.5	45.9	492.7	541.0	48.3	475.9	514.4	38.4

Table 10
Repeated Measures Analysis of Variance for SAT Scales

Effect ^a	Reading scales			
	Word study skills	Reading	Multivariate	Mathematics
Location	$p < .001$	$p < .001$	$p < .001$	$p < .01$
Race	$p < .01$	$p < .01$	$p < .05$	$p < .05$
Type	$p < .001$	$p < .001$	$p < .001$	$p < .001$
Location × Race	—	—	—	—
Location × Type	—	—	—	—
Grade	$p < .001$	$p < .001$	$p < .001$	$p < .001$
Grade × Location	$p < .001$	$p < .001$	$p < .001$	$p < .05$
Grade × Race	$p < .01$	—	$p < .05$	—
Grade × Type	$p < .01$	$p < .001$	$p < .01$	—
Grade × Location × Race	—	$p < .05$	—	—
Grade × Location × Type	$p < .05$	—	—	—

Note. Only statistically significant results presented; dash indicates statistically nonsignificant results.

^aTests of between-group effects were obtained as follows: Location differences were tested unconditionally; race and type differences tested eliminating both other main effects. Both two-way interactions were tested eliminating main effects and the other two-way interaction. Insufficient error degrees of freedom precluded testing the Race × Type or Location × Race × Type interaction, or interaction of either of these with grade.

growth by location (means not tabled) is largely attributable to smaller growth in the inner-city schools. For example, the kindergarten–Grade 1 mean difference in urban, suburban, and rural schools ranges from 84.5 to 89.9 points, but for inner-city schools it is 56.9 points. The significant Grade $\times$ Race interaction (means not tabled) is due to a greater gain, on the average, among white students—for example, 96.1 points on word study skills, compared with 67.9 points gain for minorities. Although not statistically significant, whites also gained more than minorities on both the reading and mathematics scales. The lack of significance may be explained in part by the small number of degrees of freedom for error¹⁰ and also by the significant Grade $\times$ Location $\times$ Race interaction for reading. On this scale, minorities gained more than white students only in urban schools.

Class Size and 1-Year Gains

The Grade $\times$ Type interaction for reading is indicative of significantly superior growth among students in small and teacher-aide classes. The means (Table 9) further illuminate this finding. At the end of kindergarten, regular and teacher-aide classes had very similar performance—well below that of small classes. Students in small classes already had a decided advantage in reading and mathematics. Small classes maintained and even increased their superiority over regular classes by the end of the first grade. At the same time, teacher aides had more of a distinct impact in first grade than in kindergarten. Teacher-aide classes gained as much as small classes during the year, narrowing but not closing the gap between them in final performance.

The univariate *F* ratio for the Grade $\times$ Location $\times$ Type interaction is significant for word study skills, although the multivariate test for the pair of reading scales is not. The average 1-year growth on word study skills is largest for small classes in three of the four locations (Table 11) but in rural schools it is exceeded by growth in teacher-aide classes. Also, classes with teacher aides improved more than regular classes in all but urban schools. In spite of this variability from one location to another, growth from kindergarten to first grade is greater in small than in regular size classes, and overall performance is higher in small classes than in either regular or teacher-aide classes in all locations on all three measures.

Although the longitudinal sample does not permit tests of significance for the interaction of class type with race, the mean 1-year growth is informative nevertheless (see Table 9). White students' gains in mathematics were about the same regardless of class type, whereas gains in reading were slightly greater in small and teacher-aide classes. In contrast, minority students were more differentiated by class type. The 1-year gains on word study skills and reading were substantially greater in small classes, and the 1-year gains in mathematics were notably larger in small and teacher-aide classrooms.

Table 11
One-Year Growth on Word Study Skills
by Location and Class Type

Location	Class type		
	Small	Regular	Aide
Inner city	69.6	53.2	57.8
Urban	94.9	93.9	81.2
Suburban	95.4	74.7	91.4
Rural	95.7	91.3	105.0

These outcomes may be viewed in terms of a white–minority achievement gap. In kindergarten, the race difference on word study skills is 11.0 points in small classes, 10.8 in regular classes, and 8.6 in teacher-aide classes—all about the same order of magnitude. By the end of first grade, the white–minority differential was 28.8 points for students in small classes and 41.0 and 49.4 points in regular and teacher-aide classes, respectively. The 41-point difference is closest to that which might have been expected if there were no intervention. However, with students attending small classes for 2 years, the difference was reduced to 28.8 points. This did not occur at the sacrifice of performance among whites, who also gained more in small than regular classes, but by giving an extraordinary boost to minority pupils. The same pattern is seen for reading and, to a lesser extent, for mathematics as well. It must be emphasized that these results are not confirmed by statistical tests. However, classes of 15 pupils appear to have benefited both white and minority students in terms of 1-year gains in reading and mathematics, with the greater relative benefit accruing to minorities.

Discussion

This research leaves no doubt that small classes have an advantage over larger classes in reading and mathematics in the early primary grades. In a large-scale experiment in which students were randomly assigned to treatment conditions within schools, first-grade students in smaller classes benefited in terms of improved performance on standardized reading and mathematics tests and pass rates on curriculum-based tests. Those pupils who had been in small classes for 2 years benefited in both kindergarten and Grade 1, and also showed significantly greater growth during first grade on standardized reading measures when compared with their regular class peers.

There is also some indication that regular-size classes benefit from the presence of a full-time teacher aide. In the first-grade cross-sectional analysis, teacher-aide classes outperformed regular classes on each achievement

measure, but the results were not statistically significant. In the 2-year longitudinal subsample, teacher-aide classes gained as much as small classes, from the end of kindergarten to the end of first grade, on standardized reading measures. Continuing experimentation is needed to confirm this effect.

In addition to an overall class-size effect, there is strong indication that the performance of minority students is enhanced in the small-class setting. This important outcome is statistically confirmed only in inner-city and suburban areas, but the same trend is seen in urban and rural schools as well. Also, minority students in the longitudinal subsample experienced greater relative growth than white students in the second year of small-class participation.

In the cross-sectional analysis, the differential impact of small classes on minority students is revealed especially in the curriculum-based reading and mathematics tests. These tests are tied more closely to the instructional objectives and procedures of these classrooms than are the standardized Stanford measures. If classroom interventions for low-SES, minority, or at-risk students are attempted, they must be assessed in terms of the actual content that was presented and the actual extent to which students participated in learning that material.

The outcomes of this investigation differ from those of the Shapson et al. (1980) experiment that found "no significant differences attributable to class size for art, composition, vocabulary, reading, and mathematics-problem solving" (p. 149). However, the earlier study differs from the current one in important ways. For one, students in the smallest classes (16 pupils) and the largest classes (37 pupils) were not kept in the same size class for both years of the study. Perhaps of greater importance, students in the Shapson et al. study were enrolled in third and fourth grade. It is possible, as suggested by Slavin (1989), that reducing class size can have a greater effect in earlier years, especially if those are the years during which a particular subject area is emphasized. If reading as a separate subject is emphasized in first grade, then that is where we might expect the strongest effect. Shapson et al. did find a significant impact of class size on the understanding of mathematical concepts; perhaps mathematics received greater emphasis in Grades 4 and 5. The small-class advantage occurred only in classes of 16 students, close to the size of small classes in the present study.

Although this experiment yields an unambiguous answer to the question, Is there class-size effect?, other related questions remain unanswered. The issue of whether the benefits offset the costs involved is beyond the scope of this paper. However, it should be noted that few well-defined interventions have shown as consistent an impact as this one on the performance of minority students in inner-city settings, not to mention both their minority and nonminority peers in other settings.

The question of why, or under what instructional conditions, small classes work best is a complex but particularly important issue. Three dimensions of school process should be examined. First, teachers' enthusiasm and satisfaction may be enhanced when there are fewer students to teach; this may be perceived by the students and influence their own motivation for learning. The Glass, Cahen, Smith, and Filby (1982) meta-analysis concluded that "class size affects teachers. In smaller classes, their morale is better; they like their pupils better, have time to plan, and are more satisfied with their performance" (p. 65). Teachers in the Shapson et al. (1980) experiment indicated that they were especially pleased with their smaller classes, and informal comments from teachers in the present investigation confirm this view.

Second, reduced class size may directly impact on teacher-student interactions, allowing for more individual attention in particular; third, smaller classes may increase the extent to which individual pupils attend to and become involved in learning activities.

The few studies that obtained data on pupil participation or teacher-pupil interactions found very few differences between smaller and larger classrooms. For example, of many teacher and pupil variables observed in the Shapson et al. (1980) study, only the proportion of pupils addressed by the teacher as individuals increased significantly and monotonically as classes were reduced from 37 to 30 to 23 to 16 pupils. The Tennessee project did collect teacher logs—reports of how teachers spent each 15-minute interval of a typical school day. Each time interval was coded by subject area (reading, mathematics, other subject, or non-instructional time) and by the type of instruction (large group, small group, or individualized). Although the logs may have reflected teachers' lesson plans more than their actual classroom behavior, extensive analysis failed to uncover any systematic differences among small, regular, and teacher-aide classes.¹¹

These critical questions about classroom process are not only pertinent to reduced size classes. It is clear from recent scholarly writing that students' active involvement in schooling, in various forms, is an essential requisite to the successful completion of 12 years of schooling (Ekstrom, Goertz, Pollack, & Rock, 1986; Finn, 1989; Hawkins & Lam, 1987; Wehlage, 1986). Reduced class size may be one of many ways to promote a greater variety and extent of participatory behavior. Augmenting regular classes with specialized teacher aides may be another, although we must seek to understand their roles and functions in the classroom better than we do now. In either case, it is important that the process issues—What is taught? How is it taught? How are students responding?—be addressed as well as the issue of learning outcomes.

Notes

The authors are grateful to the Project STAR directors and staff for providing the data for this investigation. Appreciation is extended to Elizabeth Word, Project Director; Helen Bain, John Folger, John Johnston, and Nan Lintz, the STAR Consortium; and DeWayne Fulton and Jayne Zaharias, data managers.

This manuscript was completed while the first author was Visiting Professor in the School of Education at Stanford University. Request for reprints should be sent to Dr. Jeremy D. Finn, Department of Counseling and Educational Psychology, State University of New York at Buffalo, 408 Baldy Hall, Buffalo, NY 14260.

¹These are studies numbered 3, 6, 7, 8, and 14 in Table A.1 of Glass, Cahen, Smith, and Filby (1982). Study number 1 has also been excluded, because of irrelevant course content (tennis).

²In May, 1985, the Tennessee legislature passed House Bill 544 authorizing and funding a major longitudinal study to consider the effects of class size on pupils in grades K–3. A four-university consortium worked with the State Department of Education to plan and execute the study, and to select an advisory committee and an external research design consultant.

³Training was instituted at the second-grade level, but does not affect the first-grade results reported here.

⁴About 10% of youngsters in Tennessee do not attend kindergarten at all.

⁵The kindergarten analysis revealed no statistically significant differences in outcome between regular and teacher-aide classes (Achilles, Bain, Bellott, Folger, Johnston, & Lintz, 1988), although teacher-aide classes scored slightly better than the regular classes on each SAT scale. Thus, the comparison of small classes with the average of the others in first grade may be somewhat conservative.

⁶Once the first-grade reassignments had been made, the percentage of minority students in small, regular, and teacher-aide classes was 32.8, 36.4, and 29.4, respectively; the percentage of students receiving free lunches was 47.5, 51.3, and 51.1, respectively. Small classes had somewhat fewer minority students and students from lower-SES homes when compared with regular classes. The difference in percentages is small, amounting to less than 1 pupil in a typical class of 15 or 22.

⁷If a class had two white students or fewer, or two minority students or fewer, it was considered to be all-minority or all-white, respectively. The one or two “other” students were eliminated from all analyses.

⁸Results not tabled.

⁹Neither the BSF nor the SCAMIN measures are scaled to permit comparisons between kindergarten and Grade 1.

¹⁰Due to the reduced sample size, no correct estimate of error variation is available to test the Race $\times$ Type or the Location $\times$ Race $\times$ Type interaction. Also, the race main effect is tested with very few error degrees of freedom; the error term is Schools $\times$ Race variation for which there are only 7 degrees of freedom. These restrictions affect the interactions of each of these terms with grade differences as well.

¹¹The Tennessee study has been collecting data on problems encountered by the teachers during the school year, and interviews each teacher at the end of the year. To date, these data have not been thoroughly analyzed. Also, a Grade-4 follow-up is planned, at which point all students will have returned to regular class settings. Questionnaires will be filled in by the fourth-grade teachers regarding the pupils’ participation and initiative taking in class, as well as end-of-year achievement scores.

References

- Achilles, C. M., Bain, H. P., Bellott, F., Folger, J., Johnston, J., & Lintz, M. N. (March 1988). *Project STAR technical report. Year one (1985–86): Kindergarten results*. Nashville, TN: State Education Department.

- Davis, T. M., & Johnston, J. M. (1987). On the stability and internal consistency of the Self-concept and Motivation Inventory: Preschool/Kindergarten form. *Psychological Reports*, 61, 871–874.
- Davis, T. M., Sellers, P. A., & Johnston, J. M. (1988). The factor structure and internal consistency of the Self-concept and Motivation Inventory: What face would you wear? Preschool/Kindergarten form. *Educational and Psychological Measurement*, 48, 237–246.
- Ekstrom, R. B., Goertz, M. E., Pollack, J. M., & Rock, D. A. (1986). Who drops out of high school and why? Findings from a national study. *Teachers College Record*, 87, 356–373.
- Finn, J. D. (1989). Withdrawing from school. *Review of Educational Research*, 59, 117–142.
- Finn, J. D., & Bock, R. D. (1985). *MULTIVARIANCE VII user's guide*. Mooresville, IN: Scientific Software.
- Glass, G. V., Cahen, L. S., Smith, M. L., & Filby, N. N. (1982). *School class size: Research and policy*. Beverly Hills: Sage.
- Glass, G. V., & Smith, M. L. (1978). *Meta-analysis of research on the relationship of class size and achievement*. San Francisco: Far West Laboratory for Educational Research and Development.
- Hawkins, J. D., & Lam, T. (1987). Teacher practices, social development and delinquency. In J. D. Burchard & S. N. Burchard (Eds.), *Prevention of delinquent behavior* (pp. 241–274). Beverly Hills: Sage.
- Milchus, N., Farrah, G., & Reitz, W. (1968). *The Self-concept and Motivation Inventory: What face would you wear?* Dearborne Heights, MI: Person-O-Metrics.
- Robinson, G. E., & Wittebols, J. H. (1986). *Class size research: A related cluster analysis for decision making*. Arlington, VA: Educational Research Service.
- Shapson, S. M., Wright, E. N., Eason, G., & Fitzgerald, J. (1980). An experimental study of the effects of class size. *American Educational Research Journal*, 17, 141–152.
- Slavin, R. E. (1989). Achievement effects of substantial reductions in class size. In R. E. Slavin (Ed.), *School and classroom organization* (pp. 247–257). Hillsdale, NJ: Erlbaum.
- Slavin, R. E., & Madden, N. A. (1987). *Effective classroom programs for students at risk* (Report No. 19). Baltimore, MD: Johns Hopkins University, Center for Research on Elementary and Middle Schools.
- Tennessee Department of Education (1987). *STAR criterion referenced test. Manual for test administration. Grade 1*. Nashville, TN: State Education Department.
- The Psychological Corporation (1985). *Stanford Achievement Test Series. Technical data report*. Orlando, FL: Harcourt, Brace, Jovanovich.
- Wehlage, G. G. (1986). At-risk students and the need for high school reform. *Education*, 107, 18–28.
- Wylie, R. C. (1979). *The self-concept* (rev. ed., Vol. 2). Lincoln: University of Nebraska Press.